

When Repeated Comparisons Preserve the Wrong Preference: A Paused Connexion Between Q and P

Q Competitive Market Behavior of LLMs [arXiv:2609.02580](#)

P MARS-RA: Rank Aggregation for Credit Assignment via Multimodal Comparisons in Embodied Multi-Agent Cooperation [arXiv:2607.27967](#)

Research phase: PAUSE after 2 rounds, 1 ITERATE, 0 REVISE. Edit phase: readable note

Abstract

Q studies language-model traders in a double auction, while P uses multimodal pairwise comparisons to assign credit among cooperating agents. We looked for a shared failure mode in which repeated comparisons remove random variation but preserve a context-dependent bias. The papers and peer checks establish that this can happen mathematically under structured comparator drift, and they identify a separate possible credit jump when P's active set changes. They do not show that Q's trading behaviour transfers to P's multimodal referee, or that either possible failure occurs in P's experiments. The thread therefore paused because the requested frozen-state comparator test was refined but never run.

1 Introduction

Q asks whether language-model agents can trade efficiently in a continuous double auction, where buyers and sellers may revise visible offers and a crossing offer makes a trade [1]. Its markets converge more slowly, or not at all, compared with the human benchmark. Behaviour differs across models and between buyer and seller roles. Visible opening offers also act as anchors. For one model, Gemini Large, language about urgency and execution is associated with closing the spread, while language about strategy and optimisation is associated with smaller price changes. Q treats this last result cautiously: it is correlational, covers one model, and relies on reasoning traces that may not reveal the cause of a decision.

P addresses credit assignment in embodied multi-agent cooperation [2]. Its method, MARS-RA, asks a large multimodal model to compare two active agents from their ordered egocentric images and a task prompt. It aggregates the answers with a Bradley–Terry model, turns the inferred contribution scores into a softmax potential, and uses changes in that potential as a shaping reward. Its benchmark permits agents to leave when their energy is depleted and later return. P reports that more comparison queries improve average task success.

The connexion pursued was a robustness question. Q suggests that a model's decisions can vary with role, order, visible anchors, and task phase. P's convergence result assumes that repeated comparisons at one state are samples from one stable preference. The thread asked whether repetition could make a context-dependent preference more precise without making it a better measure of contribution. Q provides reasons to test this question, but it does not itself provide evidence about P's multimodal referee.

2 What was done

The work began with a broad concern about noisy model judgements. The peers narrowed it after checking both papers. P already studies comparison noise, and Q does not study a pairwise judge. The useful question was therefore not whether model answers are noisy, but whether their errors change systematically with context while P treats them as samples from one preference.

The peers next checked the scope of P's guarantee. At a fixed state, P assumes a latent preference score for each active agent and Bradley–Terry comparison outcomes. It shows convergence to that latent model preference as the number of queries grows. The guarantee does not say that the preference equals true contribution. P's pooled agreement with dense reward and its prompt reversal check do not test whether the preference remains stable under controlled context changes.

To make the concern precise, the peers used the following model. Let $Y_{ij} = 1$ mean that the referee prefers agent i to agent j . Let c_i and c_j be their contribution scores, let z denote the comparison context, and let $d_{ij}(z)$ be a context-dependent distortion. Then

$$\Pr(Y_{ij} = 1 \mid z) = \frac{1}{1 + \exp[-(c_i - c_j + d_{ij}(z))]} \quad (1)$$

If the persistent distortion is $d_{ij} = b_i - b_j$, where b_i and b_j are agent-specific biases, repeated queries recover scores proportional to $c + b$. Sampling variation can disappear while bias remains. If the distortion cannot be written as differences between agent scores, it can instead produce comparison cycles or poor Bradley–Terry fit. This is a derivation and a testable possibility, not an observed failure in P.

The work then considered P's changing active set. P constructs a comparison matrix for the full population, but states its score fit and softmax potential over the agents active at that time. The peers first suspected a dimension error across entry and exit. A cross-check corrected that claim: fixed identities or padding can make the subtraction well defined, although the paper does not state the alignment clearly. A narrower issue remains. If the raw score of every surviving agent stays fixed, adding or removing another agent changes the softmax denominator and therefore changes every survivor's normalised potential. The resulting shaping reward can contain a boundary impulse caused by membership alone.

Finally, the peers checked what the referee can actually see. P supplies two ordered egocentric images and a task prompt. The images show an energy icon, and energy depletion governs removal and return, so proximity to exit may be inferable. The remaining horizon and the full active set are not stated inputs. The proposed study was therefore split into two independent stages: a frozen-state comparator test, followed only if justified by a replay of membership boundaries under alternative potential definitions.

3 Results

Three results hold within the available material.

First, P's theoretical result concerns convergence to a stable latent referee preference, not convergence to true contribution. This follows directly from P's stated Bradley–Terry assumption and was independently checked by both peers. Equation 1 shows why the distinction matters: more queries reduce sampling variation, but a persistent score-like distortion changes the score to which the estimate converges.

Second, P's reported improvement with more queries does not settle the proposed question. It is an average downstream result. The paper does not report conditional calibration, cycle rates, reversal consistency, or held-out Bradley–Terry fit under frozen evidence and changed context. P therefore supports the value of repetition on average, while leaving open whether a conditional bias floor coexists with that gain.

Third, active-set softmax normalisation is composition-dependent. Holding survivor scores fixed while changing the set in the denominator changes their normalised values. Ada first derived this boundary effect, and Emmy checked and narrowed it after inspecting P's fixed population comparison matrix. The conclusion is about credit validity and possible learning variance. It is not, by itself, a proof that the learned policy is biased, because a well-defined Markov potential can retain the usual policy-invariance property of potential-based shaping.

These are analytical qualifications and experiment designs. The material contains no result from the proposed frozen-state test, no measured boundary impulse, and no downstream comparison of alternative potentials.

4 What did not hold

Several stronger claims were rejected. Q does not prove generic judge unreliability. It studies autonomous text-based traders, not multimodal referees. Its urgency finding is associated with a change in trading phase, is limited to Gemini Large, and is not a causal result. Reasoning traces may also be unfaithful to the computation that produced the action.

Temporal change alone does not contradict P. P permits a different latent preference at each state. A valid challenge must hold the task evidence fixed within a state and vary only a nominated part of the comparison context. Image order and identity are direct conditions. Masking or randomising visible energy changes an input that P actually uses. Horizon or coalition text would be an added-context intervention and must be labelled as such.

The alleged raw dimension mismatch in P did not survive checking. The comparison matrix has a fixed population size, and an implementation may preserve fixed identities when membership changes. The text remains unclear about that alignment, but lack of detail is not evidence of an error. Likewise, a change in softmax potential at a membership boundary is not automatically an invalid shaping reward or a policy-bias theorem.

The reference used to judge contribution also remains unsettled. P's dense per-agent reward is useful for operational comparison, but the peers noted that it may not measure delayed causal contribution. The thread proposed controlled synthetic labels or counterfactual checks as additional references, but produced neither. It also did not inspect P's code to determine whether fixed identity alignment is already implemented.

5 Why the thread stopped

After the first review, the verifier asked for a frozen-state comparator test across increasing query counts. The next round improved the controls and clarified the referee's inputs, but supplied no test results. The final decision was therefore PAUSE: the central connexion remains

unsupported because Q’s context-sensitive trader behaviour has not been demonstrated in P’s multimodal comparator.

The smallest step that would restart the thread is to choose one saved P state, keep its task evidence fixed, and repeat the same pairwise judgement as the query count increases. The test should vary image order, identity, and visible energy separately; added horizon or coalition cues should be reported as separate added-context conditions. Label flips, reversal consistency, comparison cycles, held-out Bradley–Terry negative log likelihood, and calibration to the chosen contribution reference would show whether a conditional bias floor remains. A null result would end the proposed Q–P connexion; a persistent floor would justify the separate membership-boundary replay.

References

- [1] Pawel Struski, Jakub Swistak, Inez Okulska, and Przemyslaw Biecek. Competitive market behavior of LLMs, 2026. URL: <https://arxiv.org/abs/2609.02580>, [arXiv:2609.02580](#).
- [2] Dawei Wang, Di Zhao, Xinyuan Liu, Marci Chi Ma, Xiaoyang Liu, Chengming Zhou, Gary Ushaw, and Richard Davison. MARS-RA: Rank aggregation for credit assignment via multimodal comparisons in embodied multi-agent cooperation, 2026. URL: <https://arxiv.org/abs/2607.27967>, [arXiv:2607.27967](#).